Worksheet 6 solution - Gradient, Taylor series, the 2nd derivative test.

1. Four sharks are at the origin of coordinates when they smell blood in the water. The first shark swims 10 km/hr at an angle 37° east of true north and finds the concentration of blood is increasing at a rate of 20 (ml/m^3)/min. The second shark swims 15 km/hr at an angle 53° west of true north and finds the concentration is decreasing at a rate of 15 (ml/m^3)/min.

   SOLUTION:

   Let f be the concentration of blood and let $\mathbf{v}_i$ be the velocity of the ith shark. By the chain rule, the rate of change experienced by the ith shark with be $\nabla f|_{(0,0)} \cdot \mathbf{v}_i$. We know $\mathbf{v}_1 = (6,8)$ and $\mathbf{v}_2 = (-12,9)$, so we get

   $$6f_x + 8f_y = 20$$
   $$-12f_x + 9f_y = -15$$

   Solving gives $\nabla f|_{(0,0)} = (f_x, f_y) = (2,1)$.

   (a) The third shark swims southeast at 20 km/hr. At what rate does it find the concentration of blood is changing?

   SOLUTION:

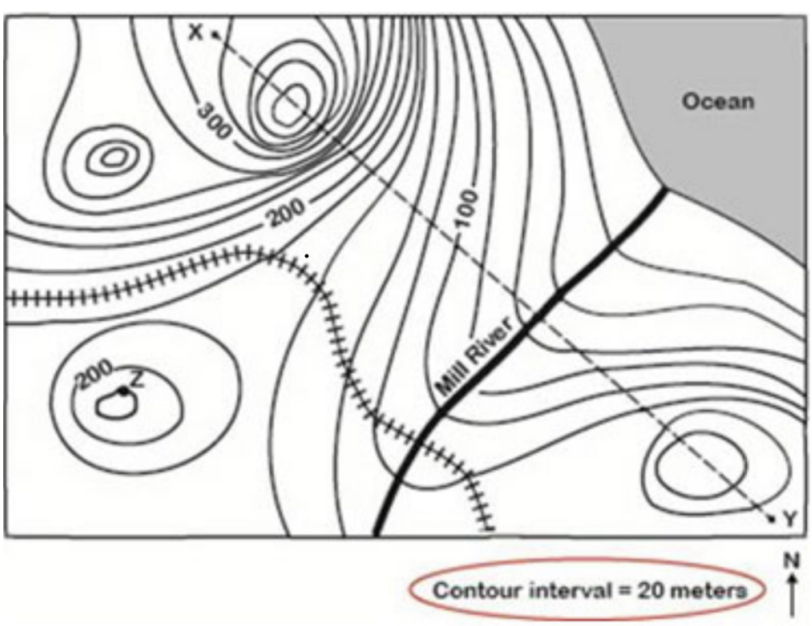
   $\mathbf{v}_3 = (10\sqrt{(}2), -10\sqrt{2})$, so the rate of change is

   $$(2,1) \cdot (10\sqrt{(}2), -10\sqrt{2}) = 10\sqrt{2} \approx 14.1(\text{ml/m}^3)/\text{min}.$$

   (b) In which direction should the fourth shark swim so as to make the concentration increase as quickly as possible?

   $\nabla f|_{(0,0)} \cdot \mathbf{v}_4 = |\nabla f|_{(0,0)}||\mathbf{v}_4|\cos\theta$ where θ is the angle between $\mathbf{v_4}$ and the gradient vector. to maximize this quantity for a given speed, we should take $\theta = 0$, i.e the shark should swim in the direction of the gradient vector $(2,1)$.

2. Consider the contour map shown below. Based on what you know about gradients (and rivers), what is the angle between the river and the contours? Why must this be the case?



SOLUTION:

The river is perpendicular to the contours.

Water will tend to flow down the hill along the path of steepest descent. The direction of $-\nabla f(a,b)$ is the direction of steepest descent at (a,b) and the gradient is perpendicular to the level curves. Putting this together, we conclude the river is perpendicular to the contours.

3. Consider $f(x,y) = 2\cos x - y^2 + e^{xy}$.

   (a) Show that $(0,0)$ is a critical point for f.

   SOLUTION:
   $\frac{\partial f}{\partial x}|_{(0,0)} = (-2\sin x + ye^{xy})|_{(0,0)} = 0$ and $\frac{\partial f}{\partial y}|_{(0,0)} = (-2y + xe^{xy})|_{(0,0)} = 0$

   (b) Calculate each of f_{xx}, f_{xy}, f_{yy} at $(0,0)$ and use this to write out the 2nd-order Taylor approximation for f at $(0,0)$.

   SOLUTION:

   The second order Taylor approximation of a function $f(x,y)$ at $(0,0)$ is given by $T_2(x,y) = f(0,0) + f_x(0,0)x + f_y(0,0)y + (f_{xx}(0,0)/2)x^2 + (f_{yy}(0,0)/2)y^2 + f_{xy}(0,0)xy$. For this problem we have $f_{xx} = -2\cos x + y^2e^{xy}$, $f_{yy} = -2 + x^2e^{xy}$, and $f_{xy} = e^{xy} + xye^{xy}$. So $f_{xx}(0,0) = -2 = f_{yy}(0,0)$ and $f_{xy}(0,0) = 1$. Also $f(0,0) = 3$. So the second order Taylor approximation for f at $(0,0)$ is $g(x,y) = 3 - x^2 - y^2 + xy$.

4. Let $g(x,y)$ be the approximation you obtained for $f(x,y)$ near $(0,0)$ in 1(b). It's not clear from the formula whether g, and hence f, has a min, max, or a saddle at $(0,0)$. Test along several lines until you are convinced you've determined which type it is. In the next problem, you'll confirm your answer in two ways.

   SOLUTION:

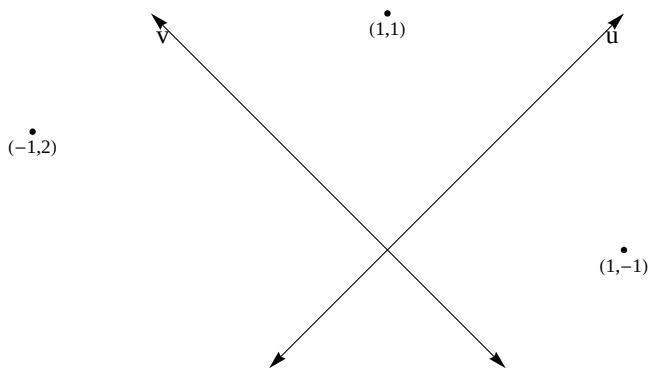
   Let's test a general line $y = mx$ which goes through $(0,0)$ as $x \to 0$. Then $g(x,mx) = 3 - x^2 - m^2x^2 + mx^2 = 3 - (1 - m + m^2)x^2$. The polynomial $1 - m + m^2$ is always positive (it opens upward and has its global minimum at $m = 1/2$ where $1 - m + m^2 > 0$). So $g(x,mx)$ is always a downward opening parabola. This suggests that $(0,0)$ is a relative maximum.

5. Consider alternate coordinates (u,v) on $\mathbb{R}^2$ given by $(x,y) = (u - v, u + v)$.

   (a) Sketch the u- and v-axes relative to the usual x- and y-axes, and draw the points whose (u,v)-coordinates are: $(-1,2)$, $(1,1)$, $(1,-1)$.

   SOLUTION:

   If we express u and v in terms of x and y we get $u = 1/2(x+y)$ and $v = 1/2(y-x)$. So the u-axis is given in x and y coordinates by all multiples of the vector $(1,1)$ and the v-axis is given by all multiples of the vector $(-1,1)$. The two axes and the points are shown below.



(b) Express g as a function of u and v, and expand and simplify the resulting expression.

SOLUTION:

$3 - x^2 - y^2 + xy = 3 - (u-v)^2 - (u+v)^2 + (u-v)(u+v) = 3 - (u^2 - 2uv + v^2) - (u^2 + 2uv + v^2) + u^2 - v^2 = 3 - u^2 - 3v^2$.

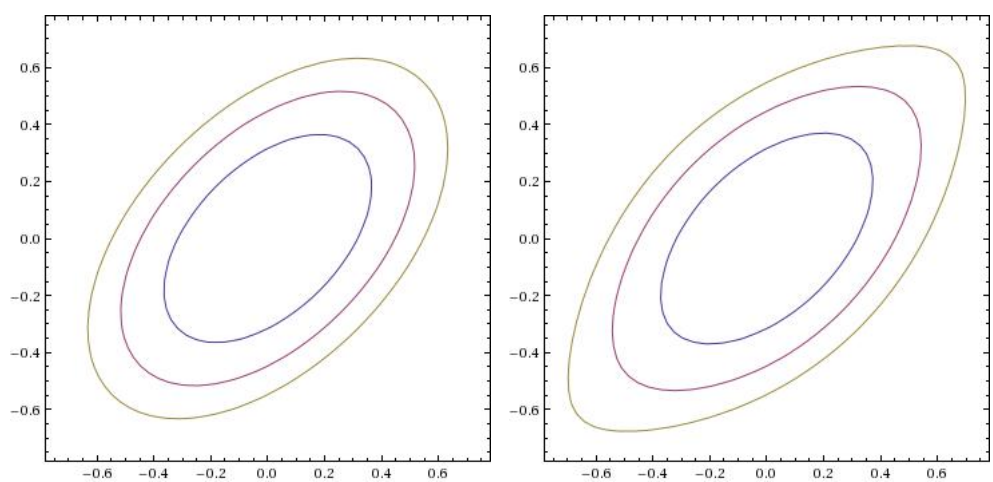
(c) Explain why your answer in 3(b) confirms your answer in 2.

SOLUTION:

This is an elliptic paraboloid (in uv coordinates) opening downward with maximum at $(0,0,3)$, so it confirms that $(0,0)$ is a local maximum ($(0,0)$ goes to $(0,0)$ under the transformation, so this reasoning makes sense).

(d) Sketch a few level sets for g. What do the level sets of f look like near $(0,0)$?

SOLUTION: The level sets are sketched for $g = 2.7, 2.8, 2.9$ on the left and for $f = 2.7, 2.8, 2.9$ on the right. The level sets for g are ellipses that approximate the level sets of f close to $(0,0)$. The ellipses shrink as they get closer to $g(x,y) = 3$, which consists of the single solution $(x,y) = (0,0)$.



(e) It turns out that there is always a similar change of coordinates so that the Taylor series of a function f which has a critical point at $(0,0)$ looks like $f(u,v) \approx f(0,0) + au^2 + bv^2$. In fact this is why the 2nd derivative test works.

Double check your answer in 2 by applying the 2nd-derivative test directly to f.

SOLUTION:

The Hessian $f_{xx}f_{yy} - (f_{xy})^2$ is $(-2)(-2) - 1^2 = 3 > 0$ at $(0,0)$ and $f_{xx}(0,0) = -2 < 0$. So f has a relative maximum at $(0,0)$ as suspected.